

AI-Assisted Hair Care Recommendation System: A Hybrid Decision Support System Using Large Language Models

Quarshona Collins

Department of Computer Science
North Carolina A&T State University
Greensboro, NC, USA
qcollins@aggies.ncat.edu

Courtney Carr

Department of Computer Science
North Carolina A&T State University
Greensboro, NC, USA
ccarr2@aggies.ncat.edu

Abstract—This paper covers the design and development of an AI-Assisted Hair Care Recommendation System. The system combines a rule-based processing module we call the “Mini Brain” with a Large Language Model (LLM) to give users personalized hair care advice. When someone fills out our survey, the system reads their answers, figures out what hair issues they have, and sends all of that to GPT-4o-mini. The AI then comes back with a full routine, product suggestions, ingredients to look for, things to avoid, and an explanation of why it made those choices. We built the interface using Streamlit and connected it to the OpenAI API. This paper is split into two phases: Phase 1 covers the problem we identified, our research, and our first working prototype. Phase 2 goes deeper into the system design, adds UML diagrams, and shows our improved version of the app.

Index Terms—Recommender Systems, Large Language Models, Natural Language Processing, Personalized Hair Care, Decision Support Systems, Explainable AI, Hybrid Systems

— PHASE 1: REQUIREMENTS, LITERATURE & INITIAL PROTOTYPE —

I. INTRODUCTION

Walk into any beauty store and you will find hundreds of hair care products on the shelf, all promising different things. For most people, picking the right ones is basically a guessing game. You try something, it does not work, you waste money, and sometimes your hair ends up worse than before. Getting real advice from a dermatologist or hair specialist is expensive and not realistic for most people.

That is the problem we wanted to solve. We built an AI system that asks users a few simple questions about their hair and gives back a personalized routine and product recommendations, along with an explanation of why those things were suggested. The whole idea is to make professional-level hair advice accessible to anyone with a phone or laptop.

Our system pulls from research in recommendation systems [1], [2], AI personalization [3], modern language mod-

els [4], and explainable AI [5] to make sure the advice it gives is actually grounded in real methods and not just made up.

II. PROBLEM STATEMENT

A lot of people struggle to find the right hair products because their hair is complicated. Someone might have curly, low-porosity hair that is also color-treated and dry — and finding products that work for all of those things at once is really hard. Searching online usually just gives you generic tips that do not account for your specific situation.

Some of the biggest problems we saw:

- People waste a lot of money buying products that end up not working
- Using the wrong products can actually damage your hair over time
- There is no easy starting point for someone who just wants a simple routine
- A lot of the advice online is either too vague or contradicts itself

We wanted to build something that actually listens to what you tell it about your hair and gives you a real, personalized answer instead of the same generic advice everyone else gets.

III. RESEARCH OBJECTIVES

Here is what we set out to do with this project:

- 1) Build a questionnaire that collects the right information about someone’s hair
- 2) Use an AI model to understand and interpret what the user tells us
- 3) Figure out what hair conditions or issues the user has
- 4) Recommend specific product types and ingredients based on those conditions
- 5) Give the user a step-by-step routine they can actually follow
- 6) Make sure the system explains *why* it made each recommendation

IV. JUSTIFICATION FOR LLM-BASED APPROACH

We could have built this with a giant list of if-then rules — like, “if hair is coily AND dry AND chemically treated, then recommend X.” But that gets complicated really fast. Hair care involves so many combinations of conditions that you would need thousands of rules just to cover the most common situations, and even then you would miss a lot.

Large Language Models like GPT-4o-mini are much better at handling that kind of complexity. They can read a full description of someone’s hair situation and reason through it the same way a knowledgeable person would. Instead of just matching to a rule, the AI actually thinks about what would make sense for that person.

We still kept a rule-based piece in the system — our “Mini Brain” module — to handle basic input checking and flag common conditions before the AI even sees the input. That way, the AI is not starting from scratch every time and the output is more consistent. This kind of hybrid approach is backed up by research in clinical AI systems [3] where combining rules with AI gives better results than using either one alone.

V. PRELIMINARY LITERATURE REVIEW

A. Recommendation Systems

Roy and Dutta (2022) looked at how recommendation systems work across different fields and found that the best ones use detailed information about the user, not just generic preferences [1]. That is exactly why we built a questionnaire instead of just letting users type in a free-form question — we wanted structured, useful input.

B. Deep Learning-Based Recommenders

Zhang et al. (2019) showed that AI models can take in multiple features about a user at once and combine them to make better recommendations [2]. In our case, those features are things like hair type, scalp condition, and whether the user uses heat tools. The LLM handles all of those together instead of looking at them one by one.

C. NLP Personalization in Clinical Decision Support

Alharbi et al. looked at how NLP can be used in healthcare to give people more personalized advice [3]. Even though they were working in a medical context, the same idea applies here — the more the system understands about the person, the better the recommendation it can give.

D. BERT and Transformer Foundation Models

Devlin et al. (2019) introduced BERT, which was one of the first models to really show how powerful transformer-based AI could be at understanding language [4]. GPT-4o-mini, which powers our system, is built on the same kind of foundation and is what allows it to understand complex hair descriptions and respond intelligently.

E. Explainable AI for Decision Support

Holzinger et al. (2019) made the point that AI systems people actually trust need to be able to explain themselves [5]. We took that seriously — our system does not just give you a list of products, it tells you exactly why it is recommending them based on what you said about your hair.

VI. METHODOLOGY

We followed an Agile approach, working in short sprints so we could keep testing and improving as we went. Here is how we broke it down:

- **Sprint 1:** Designed the questionnaire and started mapping out the system with use case and DFD diagrams
- **Sprint 2:** Wrote and tested different versions of the AI prompt to see which gave the best results
- **Sprint 3:** Built the Streamlit interface and connected everything to the OpenAI API
- **Sprint 4 (13-day testing phase):** Tested the full system and evaluated how well it personalized, explained, and presented recommendations

The basic flow of how the system works is: the user fills out the survey, the Mini Brain scans their answers for key conditions, those conditions get packaged into a structured prompt, and the AI returns a full recommendation. We did not need any outside datasets since the AI generates everything on the fly based on what the user tells it.

VII. SYSTEM DESIGN

The system is built in three main parts that each have a specific job: the Streamlit front-end that the user interacts with, the Python backend that processes the input and builds the prompt, and GPT-4o-mini that actually generates the recommendation. Keeping these three parts separate made the system easier to build, test, and update.

A. Functional Decomposition

Fig. 1 shows how the system breaks down into smaller pieces. At the top level, the system does three things: takes in user input, builds a response, and delivers that response to the user. Each of those breaks down further — for example, “Take User Input” includes verifying that the input is valid before anything else happens.

3. FUNCTIONAL DECOMPOSITION DIAGRAM

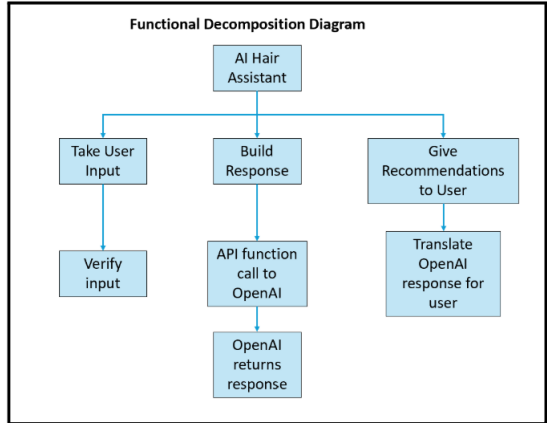


Fig. 1. Functional Decomposition Diagram of the AI Hair Assistant System

B. Data Flow Diagrams

Fig. 2 shows three levels of data flow through the system. The top-level context diagram just shows the user sending input and getting recommendations back. The Level 0 DFD breaks that into the three main processes. The lower-level DFD zooms in on the “Build Response” step and shows how the system constructs the prompt and calls the OpenAI API.

4. CONTEXT DIAGRAM, 0 DFD, AND LOWER LEVEL DFD

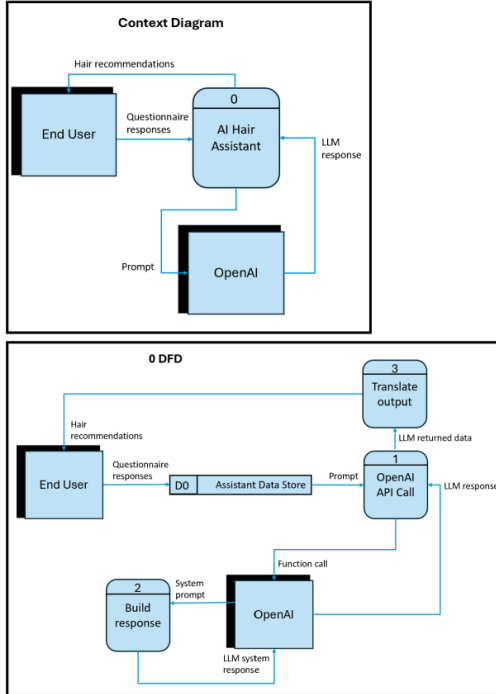


Fig. 2. Context Diagram (top), Level 0 DFD (middle), and Lower-Level DFD (bottom)

C. Decision Table — Input Verification

Before anything gets sent to the AI, the Mini Brain checks whether the input is actually usable. Table I shows the logic

behind that check — there are three conditions, and depending on which ones are true, the input is either accepted or rejected.

TABLE I
DECISION TABLE — INPUT VERIFICATION LOGIC

Verify User Input	1	2	3	4	5	6	7	8
Dropdown menus filled out	Y	–	–	Y	–	Y	N	–
Description typed in text box	–	Y	–	Y	–	–	Y	–
Input is about hair condition	N	N	Y	Y	N	Y	Y	N
Accept input			X	X		X	X	
Reject input	X	X			X			X

VIII. LLM INTEGRATION AND PROMPT ENGINEERING

The way we talk to the AI matters a lot. If the prompt is vague, the output will be vague. So we put a lot of thought into how to structure it. Every prompt has four parts: we tell the AI what role it is playing (a hair care advisor), we give it all of the user’s information, we include any flags the Mini Brain detected, and we tell it exactly what format to respond in.

A. Fine-Tuning Strategy

We used GPT-4o-mini straight out of the box — no custom training. We looked into fine-tuning but decided against it for a few reasons: there is no publicly available labeled dataset for hair care recommendations, the pre-trained model already performed really well in our tests, and training a custom model would have taken way more time than we had. In the future, we could look at using retrieval-augmented generation (RAG) to pull from a real product database.

B. Prompt Template

Here is the exact prompt structure we used:

```

prompt = f"""
You are a professional hair care advisor.
Hair Type: {hair_type}
Scalp Condition: {scalp_condition}
Hair Goal: {hair_goal}
Heat Styling Frequency: {heat_usage}
Chemical Treatments: {chemical_treatments}
Additional Description: {user_description}
Based on these inputs, provide:
1. Daily and Weekly Routine
2. Best Product Types to Use
3. Key Ingredients to Look For
4. Things to Avoid
5. Explanation: Why I Recommend This
"""
  
```

IX. PROTOTYPE IMPLEMENTATION — INITIAL VERSION

A. Initial (Original) Prototype

The first version of the app was mostly about proving the concept worked. The user picked answers from dropdown

menus, hit a button, and got a recommendation back from the AI. It was pretty bare-bones but it showed us the core idea was solid. The screenshots below are from that original version.

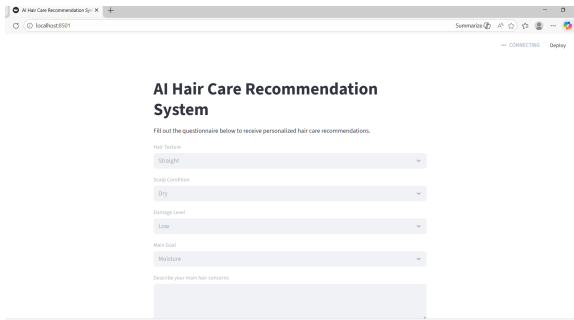


Fig. 3. Initial Prototype — Questionnaire Interface

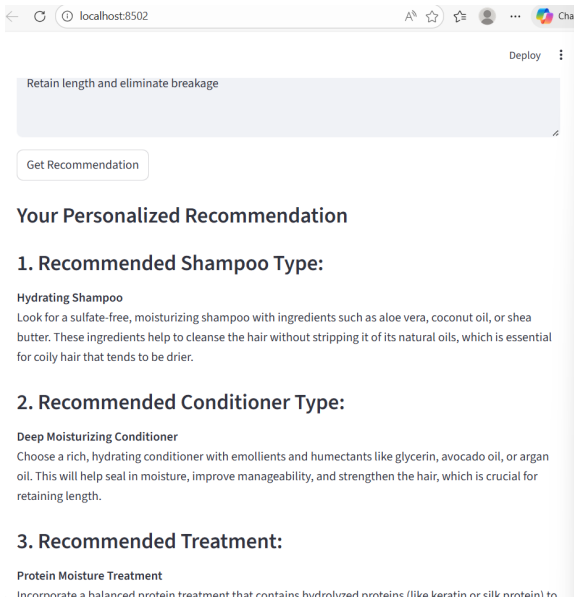


Fig. 4. Initial Prototype — Recommendation Output (Part 1)

3. Recommended Treatment:

Protein Moisture Treatment
Incorporate a balanced protein treatment that contains hydrolyzed proteins (like keratin or silk protein) to help strengthen and reinforce hair structure while maintaining moisture. Look for treatments that add hydration alongside protein to avoid making the hair too stiff.

4. Suggested Weekly Routine:

- **Day 1:**
 - Wash with hydrating shampoo.
 - Condition with deep moisturizing conditioner.
 - Apply a leave-in conditioner for added moisture.
- **Day 2:**
 - Refresh curls with a water-based curl cream or spray mixed with a little leave-in conditioner.
- **Day 3:**
 - No wash; consider a scalp massage with light oil (like jojoba) to maintain scalp health.
- **Day 4:**
 - Use a protein moisture treatment as needed (once every 3-4 weeks, not on the same week as a *deep conditioning treatment*).

Fig. 5. Initial Prototype — Recommendation Output (Part 2)

- Use a protein moisture treatment as needed (once every 3-4 weeks, not on the same week as a deep conditioning treatment).
- **Day 5:**
 - Co-wash with a cleansing conditioner (sulfate-free) to remove product build-up without stripping moisture.
- **Day 6/7:**
 - Apply a hydrating hair mask and leave on for a few hours or overnight, then rinse out and follow with conditioner.

5. Explanation for Recommendations:

Hair Type: Coily
Coily hair is typically dry and prone to breakage, making moisture retention critical. Hydrating shampoo and deep conditioners are essential to cleanse the hair without stripping natural oils.

Scalp Condition: Normal
With a normal scalp condition, you don't need specialized products for scalps dealing with oil or flakes, allowing focus on the hair's moisture needs.

Damage Level: Moderate
Since there is moderate damage, the goal should be a combination of strengthening and hydrating products. The recommended protein moisture treatment aids in reinforcing the hair structure while nourishing it.

Goal: Moisture, Retain Length, Eliminate Breakage

Fig. 6. Initial Prototype — Weekly Routine Section

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Damage Level: Moderate

Since there is moderate damage, the goal should be a combination of strengthening and hydrating products. The recommended protein moisture treatment aids in reinforcing the hair structure while nourishing it.

Goal: Moisture, Retain Length, Eliminate Breakage

The routine emphasizes moisture with essential hydrating products, which are critical for coily hair. Regular protein treatments promote strength and minimize breakage, addressing your need to retain length.

Incorporating these recommendations into your hair care routine can significantly enhance moisture retention and mitigate breakage, supporting overall hair health and growth.

Fig. 7. Initial Prototype — Explanation Section

— PHASE 2: SYSTEM DESIGN, DEVELOPMENT & ENHANCED PROTOTYPE —

X. FULL SYSTEM DESIGN — PHASE 2

Phase 2 is where we got into the formal system design work. We created all the UML diagrams, finalized the architecture, and updated the prototype based on what we learned from testing. This section walks through each of the design artifacts we produced.

A. Object Relationship Diagram

The overall system is organized into three layers, each with a specific job. Fig. 8 shows how these layers connect to each other.

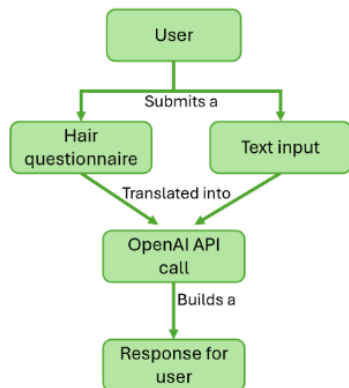


Fig. 8. System Architecture Diagram — Three-Tier Design

TABLE II
SYSTEM ARCHITECTURE LAYERS

Layer	Component	What It Does
Presentation	Streamlit UI	Shows the survey, collects answers, displays the recommendation
Processing	Python Backend + Mini Brain	Validates input, detects hair conditions, builds the AI prompt
Intelligence	GPT-4o-mini (OpenAI)	Reads the prompt and generates the personalized recommendation

B. Use Case Diagram

The Use Case Diagram (Fig. 9) shows everything a user can do with the system and how those actions connect to what happens on the back end.

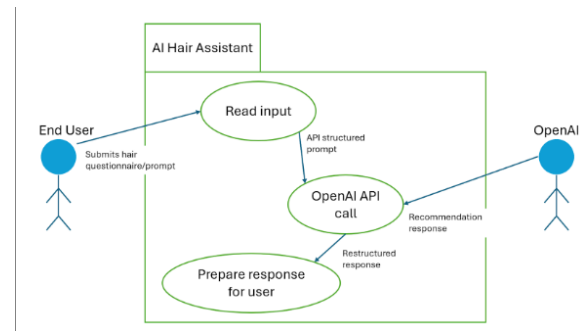


Fig. 9. Use Case Diagram — System Functionalities

C. Class Diagram

The Class Diagram (Fig. 10) shows the main objects in the system, what information they hold, and what they can do.

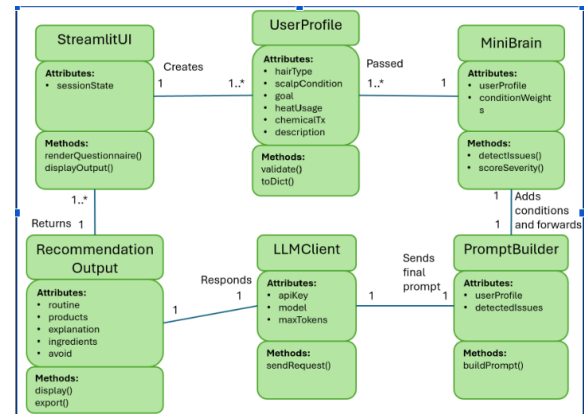


Fig. 10. Class Diagram — Key Entities, Attributes, and Relationships

D. E-R Diagram

The Entity-Relationship Diagram (Fig. 11) shows how the data in the system is organized and how the different pieces relate to each other.

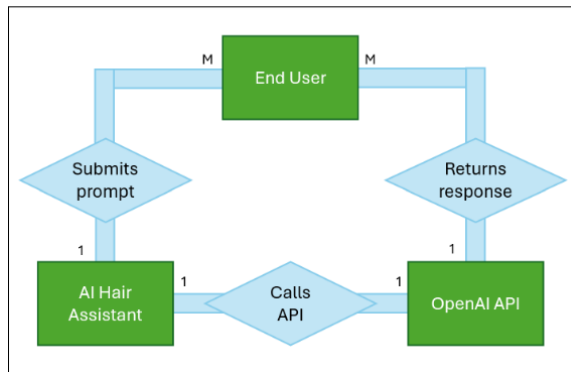


Fig. 11. Entity-Relationship Diagram

E. Extended Decision Table

Table III shows how specific combinations of hair type, scalp condition, and goal map to different recommendation priorities.

TABLE III
EXTENDED DECISION TABLE — INPUT CONDITIONS TO
RECOMMENDATION FOCUS

Hair Type	Scalp	Goal	Recommendation Focus
Curly	Dry	Moisture	Deep conditioning, leave-in cream, avoid sulfates
Straight	Oily	Frizz control	Clarifying shampoo, light serum, volumizing routine
Coily	Sensitive	Hair growth	Scalp treatment, protein-free moisturizers, gentle cleanse
Wavy	Normal	Damage repair	Protein treatment, bond repair serum, heat protectant
Any + Chemical Tx	Any	Any	Bond-building products, reduced heat, strengthening masks

XI. ENHANCED PROTOTYPE — PHASE 2

After testing the first version and getting some feedback, we went back and made a lot of improvements. The biggest changes were adding the optional text box so users can describe their hair in their own words, cleaning up the layout, and restructuring the AI output into five clearly labeled sections so it is easier to read and follow.

AI Hair Care Recommendation System

Decision Support System Prototype

Hair Assessment Survey

What is your hair type?

Straight

How would you describe your scalp?

Dry

What is your main hair goal?

Hair growth

How often do you use heat styling tools?

Never

Fig. 12. Updated Survey — Top Section

What is your main hair goal?

Hair growth

How often do you use heat styling tools?

Never

Do you use chemical treatments (relaxer, color, etc.)?

No

Tell us more about your hair (optional):

Example: My hair breaks easily at the ends and I want it longer...

Get Recommendation

Fig. 13. Updated Survey — Bottom Section with Optional Description Box

AI Hair Recommendation

Recommendations for Your Curly Hair

1. Daily and Weekly Routine

- **Daily Routine:**
 - **Moisturizing:** Use a hydrating leave-in conditioner every day to help maintain moisture levels. Be sure to apply it evenly, focusing on the mid-lengths and ends of your hair.
 - **Styling:** When styling, utilize a lightweight curl cream or gel to define curls without weighing your hair down. Avoid applying products to the roots.
- **Weekly Routine:**
 - **Deep Conditioning:** Once a week, incorporate a deep conditioning treatment or a mask specifically designed for moisture restoration and repair. Apply this treatment after shampooing and leave it in for the recommended time.
 - **Gentle Cleansing:** Adjust your cleansing routine to use a mild sulfate-free shampoo once or twice a week to remove product buildup without stripping moisture. On non-shampoo days, co-wash (use a conditioner to cleanse).

2. Best Product Types to Use

Fig. 14. Updated Output — Section 1: Daily and Weekly Routine

- **Excessive Heat:** Try to limit direct heat exposure, and consider using a heat protectant when styling with heat tools.

5. Explanation: Why I Recommend This

Your curly hair type, combined with its fine texture and heat styling habits, requires a careful balance of moisture retention and volume enhancement. By incorporating a routine that emphasizes hydration, gentle cleansing, and the right product types, you can restore moisture levels and prevent heat damage. Key ingredients like hyaluronic acid and aloe vera are ideal for fine curls, as they hydrate without adding heaviness, ensuring your curls remain lively and bouncy. Avoiding harsh sulfates and heavy products will further help maintain moisture and improve the overall health and density of your hair.

Feel free to reach out if you have any more questions or if you'd like specific product recommendations! Happy hair journey!

Fig. 16. Updated Output — Section 5: Explanation

Here is a summary of what changed between the two versions:

- Added an optional free-text box so users can give extra context in their own words
- Added the Mini Brain module to pre-process inputs before they reach the AI
- Restructured the AI output into five numbered sections that are much easier to read
- Cleaned up the overall layout so the app looks more polished
- Improved the prompt so the AI responses are more detailed and better organized

— PHASE 3: FINAL IMPLEMENTATION, EVALUATION & FUTURE WORK —

2. Best Product Types to Use

- **Leave-In Conditioner:** Look for a product designed for fine, curly hair to avoid build-up while maintaining moisture.
- **Curl Cream:** Choose lightweight, hydrating curl creams that enhance curls without the heaviness.
- **Deep Conditioning Mask:** A moisture-rich mask that targets hydration and repair is crucial.

3. Key Ingredients to Look For

- **Hyaluronic Acid:** Known for its intense moisture-retaining properties.
- **Aloe Vera:** Provides moisture and nourishment while being light enough for fine hair.
- **Shea Butter or Coconut Oil:** Foster hydration but ensure they are in lighter formulations to prevent weighing down your curls.
- **Glycerin:** A humectant that helps draw moisture into your hair.

4. Things to Avoid

- **Sulfates:** Avoid shampoos containing sulfates, as they can strip essential moisture.
- **Heavy Oils or Butters:** Products with heavy oils can weigh down fine hair and lead to greasy buildup.
- **Alcohols:** Steer clear of drying alcohols found in styling products, as they can cause dehydration.
- **Excessive Heat:** Try to limit direct heat exposure, and consider using a heat protectant when styling with heat tools.

Fig. 15. Updated Output — Sections 2–4: Products, Ingredients, Things to Avoid

XII. FINAL SYSTEM ENHANCEMENTS — PHASE 3

Phase 3 focused on expanding the system into a fully featured, production-quality prototype. The major upgrades include a significantly longer 13-question survey, an enhanced dual Mini Brain architecture with weighted scoring, a visual Hair Profile Score Dashboard, structured AI output with day-by-day routines, confidence scoring, and a user feedback loop.

The final questionnaire now collects detailed information on hair porosity, density, length, wash frequency, damage level, environment, styling habits, diet & water intake, and an optional free-text description. This richer input allows for much more accurate personalization.

The Mini Brain was upgraded with both a simple rule-based detector and an enhanced weighted scoring engine that calculates four key metrics (Moisture Level, Damage Level, Scalp Health, and Breakage Risk) displayed as a clear dashboard with progress bars. Combined condition flags were also added for high-risk situations.

After preprocessing, a carefully structured prompt is sent to GPT-4o-mini, which returns a clean recommendation organized into five clearly labeled sections. The interface includes visual confidence scoring and a 1–5 star feedback loop.

The screenshots below show the final v4.0 system:

AI Hair Care Recommendation System

Decision Support System Prototype - v4.0

Hair Assessment Survey

What is your hair type?

Straight

What is your hair density?

Fine

How would you describe your scalp?

Dry

How often do you use heat styling tools?

Never

What is your hair porosity?

Low

Do you use chemical treatments (relaxer, color, etc.)?

No

What is your current hair length?

Medium (shoulder length)

How would you rate your current damage level?

Low

How often do you wash your hair?

Daily

What type of environment do you live in?

Humid

What is your main hair goal?

Strengthen hair

Fig. 17. Final v4.0 Hair Assessment Survey (Top Section)

What is your main hair goal?

Strengthen hair

What is your most common styling method?

Protective styles (braids, twists, wigs)

How would you describe your water intake and diet?

Healthy diet, drink plenty of water

Tell us more about your hair (optional):

my edges are thin

Get Recommendation

Your Personalized Recommendation

Hair Profile Score Dashboard

Generated by Mini Brain rule-based analysis - before AI processing

Moisture Level Damage Level Scalp Health Breakage Risk

Fig. 18. Final v4.0 Hair Assessment Survey (Bottom Section)

Your Personalized Recommendation

Hair Profile Score Dashboard

Generated by Mini Brain rule-based analysis - before AI processing



Hair Health Score

Score: 6 — Higher = more damage/attention needed

Moderate condition — balanced recommendations

Detected Issues

Fig. 19. Final Hair Profile Score Dashboard

Detected Issues

Dry Scalp, Low Porosity, Fine Hair

Mini Brain — Detailed Issue Breakdown

Rule-based analysis completed before AI processing

- Dry Scalp
- Low Porosity — Product Buildup Prone
- Fine Hair — Prone to Breakage
- Low Damage Level (user-reported)
- Humid Climate — Frizz Risk

Fig. 20. Final Recommendation - Suggested Weekly Routine

AI Recommendation

Sure, here's a professional recommendation tailored to your hair type and goals:

- Recommended Shampoo Type: Hydrating Shampoo**
Look for shampoos that contain ingredients such as aloe vera, glycerin, and natural oils (like avocado or coconut oil). These ingredients provide moisture without weighing down fine hair, which is essential for low porosity hair that can struggle to absorb hydration. Hydrating shampoos help cleanse the scalp and hair while maintaining moisture levels, which is crucial for your dry scalp and overall hair health.
- Recommended Conditioner Type: Lightweight Leave-In Conditioner**
Select a leave-in conditioner formulated with ingredients like shea butter, argan oil, and hydrolyzed proteins. These ingredients provide essential moisture and help to strengthen the hair without causing buildup common with low porosity hair. The lightweight formula is perfect for fine hair, ensuring it doesn't feel heavy or greasy while promoting soft, manageable strands.
- Recommended Treatment: Protein Moisture Treatment**
A protein moisture treatment is ideal for your hair as it balances protein and moisture levels, strengthening fine hair prone to breakage. Look for treatments with hydrolyzed proteins and humectants, which can improve the hair's elasticity and strength. Because you mentioned a low damage level, a treatment used sparingly can help maintain and enhance the overall integrity of your hair.
- Suggested Weekly Routine:**
 - Day 1: Wash hair with hydrating shampoo, follow with a lightweight leave-in conditioner, style in protective braids.
 - Day 2: Moisturize scalp and edges lightly with natural oils (like castor or jojoba) to nourish and strengthen.
 - Day 3: Apply the protein moisture treatment as directed, focusing on mid-lengths to ends. Rinse thoroughly and follow with a lightweight leave-in.
 - Day 4: Rest day; wear hair in a loose protective style and hydrate with a spray of water mixed with a few drops of oil.
 - Day 5: Wash hair with hydrating shampoo, condition with a lightweight conditioner, and style as desired.
 - Day 6/7: Focus on scalp care—apply a deep moisturizing scalp treatment, and do a gentle scalp massage to promote circulation.

Fig. 21. Final Recommendation - Products and Ingredients

- Day 1: Wash hair with hydrating shampoo, follow with a lightweight leave-in conditioner, style in protective braids.
 - Day 2: Moisturize scalp and edges lightly with natural oils (like castor or jojoba) to nourish and strengthen.
 - Day 3: Apply the protein moisture treatment as directed, focusing on mid-lengths to ends. Rinse thoroughly and follow with a lightweight leave-in.
 - Day 4: Rest day; wear hair in a loose protective style and hydrate with a spray of water mixed with a few drops of oil.
 - Day 5: Wash hair with hydrating shampoo, condition with a lightweight conditioner, and style as desired.
 - Day 6/7: Focus on scalp care—apply a deep moisturizing scalp treatment, and do a gentle scalp massage to promote circulation.
- Explanation for Recommendations: Hair Type: Straight** — Products that provide hydration without heaviness are essential for straight fine hair, allowing for a sleek appearance without flatness.
- Scalp Condition: Dry** — Hydrating products are vital in alleviating dryness, ensuring the scalp remains healthy and fosters hair growth.
- Damage Level: Low** — Since your hair has a low damage level, treatments like the protein moisture treatment should be used sparingly for maintenance without overwhelming your strands.
- Goal: Strengthen hair** — All products are selected to promote moisture retention and strength, addressing both fine texture and low porosity tendencies.
- In summary, this tailored hair care regimen will cater to your specific needs and goals, while supporting the health and strength of your hair in a humid environment.



Fig. 22. Final Recommendation - Explanation Section

The complete source code for this final version is available in the project repository (file: `app.py`) and is also reproduced in full in Section XIII.

XIII. CODE DOCUMENTATION

The final application is contained in a single `app.py` file and includes the expanded questionnaire, dual Mini Brain functions, Hair Profile Score Dashboard, structured output parsing, and sidebar documentation of the full system process. The complete source code is provided below.

Demo Video: A recorded demonstration of the final v4.0 system is available at: <https://drive.google.com/file/d/1Fqz2u9xVQ9YAS1ODEFZOdeCQiz22qpVr/view>


```

1 import streamlit as st
2 import openai
3
4 # =====
5 # PAGE CONFIG
6 # =====
7 st.set_page_config(
8     page_title="AI Hair Care System",
9     page_icon="",
10    layout="wide"
11 )
12
13 st.title(" AI Hair Care Recommendation System")
14 st.subheader("Decision Support System Prototype - v4.0")
15
16 # =====
17 # API KEY
18 # =====
19 api_key = "YOUR_API_KEY_HERE"
20
21 # =====
22 # USER INPUT SECTION
23 # =====
24 st.header("Hair Assessment Survey")
25
26 col1, col2 = st.columns(2)
27
28 with col1:
29     hair_type = st.selectbox(
30         "What is your hair type?",
31         ["Straight", "Wavy", "Curly", "Coily"]
32     )
33     scalp_condition = st.selectbox(
34         "How would you describe your scalp?",
35         ["Dry", "Oily", "Normal", "Sensitive"]
36     )
37     porosity = st.selectbox(
38         "What is your hair porosity?",
39         ["Low", "Medium", "High"]
40     )
41     hair_length = st.selectbox(
42         "What is your current hair length?",
43         ["Short (ear length or above)",
44          "Medium (shoulder length)",
45          "Long (past shoulder)",
46          "Very Long (past chest)"]
47     )
48     wash_frequency = st.selectbox(
49         "How often do you wash your hair?",
50         ["Daily", "Every 2-3 days",
51          "Once a week", "Every 2 weeks or less"]
52     )
53
54 with col2:
55     density = st.selectbox(
56         "What is your hair density?",
57         ["Fine", "Medium", "Thick"]
58     )
59     heat_usage = st.selectbox(
60         "How often do you use heat styling tools?",
61         ["Never", "Sometimes", "Often"]
62     )
63     chemical_treatments = st.selectbox(
64         "Do you use chemical treatments?",
65         ["No", "Occasionally", "Frequently"]
66     )
67     damage_level = st.selectbox(
68         "How would you rate your current damage level?",
69         ["None", "Low", "Moderate", "High"]
70     )
71     environment = st.selectbox(
72         "What type of environment do you live in?",
73         ["Humid", "Dry", "Cold", "Mixed/Moderate"]
74     )
75
76 hair_goal = st.selectbox(
77     "What is your main hair goal?",
78     ["Hair growth", "Moisture", "Damage repair",
79      "Frizz control", "Scalp health",
80      "Length retention", "Strengthen hair"]
81 )
82
83 styling_habits = st.selectbox(
84     "What is your most common styling method?",
85     ["Protective styles (braids, twists, wigs)",
86      "Heat styling (flat iron, blow dryer)",
87      "Air dry / wash and go",
88      "Roller sets / flexi rods",
89      "No particular style"]
90 )
91
92 diet_water = st.selectbox(
93     "How would you describe your water intake and diet?",
94     ["Healthy diet, drink plenty of water",
95      "Average diet, moderate water",
96      "Poor diet, low water intake",
97      "Prefer not to say"]
98 )
99
100 user_description = st.text_area(
101     "Tell us more about your hair (optional):",
102     placeholder="Example: My hair breaks easily at the ends
103     ..."
104 )
105
106 # =====
107 # MINI BRAIN v2.0
108 # =====
109 def detect_hair_issues(scalp, heat, chemical, porosity,
110                        density):
111     issues = []
112     confidence = 70
113     if scalp == "Dry":
114         issues.append("Dry Scalp")
115         confidence += 10
116     if scalp == "Oily":
117         issues.append("Oily Scalp")
118         confidence += 10
119     if heat == "Often":
120         issues.append("Heat Damage")
121         confidence += 15
122     if chemical == "Frequently":
123         issues.append("Chemical Damage")
124         confidence += 15
125     if porosity == "Low":
126         issues.append("Low Porosity")
127         confidence += 10
128     if density == "Fine":
129         issues.append("Fine Hair")
130         confidence += 8
131     confidence = min(confidence, 95)
132     issues_str = ", ".join(issues) if issues else "No major
133     issues detected"
134     return issues_str, confidence
135
136 # =====
137 # MINI BRAIN v3/v4 Enhanced
138 # =====
139 def run_mini_brain(scalp, heat, chemical, porosity,
140                    density, hair_type, goal, damage,
141                    environment):
142     issues = []
143     confidence = 70
144     simple_score = 0
145
146     if scalp == "Dry":
147         issues.append("Dry Scalp")
148         confidence += 10
149         simple_score += 2
150     elif scalp == "Oily":
151         issues.append("Oily Scalp / Product Buildup Risk")
152         confidence += 10
153         simple_score += 1
154     elif scalp == "Sensitive":
155         issues.append("Scalp Sensitivity")
156         confidence += 6
157         simple_score += 1
158
159     if heat == "Often":
160         issues.append("Heat Damage Risk")
161         confidence += 15
162         simple_score += 3
163     elif heat == "Sometimes":
164         issues.append("Moderate Heat Exposure")
165         confidence += 5
166         simple_score += 1

```

```

163
164 if chemical == "Frequently":
165     issues.append("Chemical Damage")
166     confidence += 15
167     simple_score += 3
168 elif chemical == "Occasionally":
169     issues.append("Mild Chemical Processing")
170     confidence += 8
171     simple_score += 1
172
173 if porosity == "Low":
174     issues.append("Low Porosity - Product Buildup Prone")
175     confidence += 10
176     simple_score += 1
177 elif porosity == "High":
178     issues.append("High Porosity - Moisture Loss Risk")
179     confidence += 10
180     simple_score += 2
181
182 if density == "Fine":
183     issues.append("Fine Hair - Prone to Breakage")
184     confidence += 8
185     simple_score += 1
186
187 if damage == "High":
188     issues.append("High Damage Level (user-reported)")
189     confidence += 10
190     simple_score += 3
191 elif damage == "Moderate":
192     issues.append("Moderate Damage Level (user-reported)")
193     confidence += 5
194     simple_score += 2
195 elif damage == "Low":
196     issues.append("Low Damage Level (user-reported)")
197     simple_score += 1
198
199 if environment == "Dry":
200     issues.append("Dry Climate - Increased Moisture Loss")
201     simple_score += 1
202 elif environment == "Humid":
203     issues.append("Humid Climate - Frizz Risk")
204     simple_score += 1
205
206 if goal == "Damage repair":
207     issues.append("Existing Damage (user-reported goal)")
208     simple_score += 2
209 elif goal == "Length retention":
210     issues.append("Length Retention Goal")
211     simple_score += 1
212
213 # Combined flags
214 if scalp == "Dry" and heat == "Often":
215     issues.append("HIGH DAMAGE RISK: Dry scalp + frequent heat")
216     confidence += 5
217     simple_score += 2
218 if chemical == "Frequently" and porosity == "High":
219     issues.append("SEVERE DAMAGE: Chemical overprocessing + high porosity")
220     confidence += 5
221     simple_score += 2
222 if scalp == "Oily" and chemical == "Frequently":
223     issues.append("Scalp imbalance: Oily scalp + chemical treatments")
224     confidence += 3
225     simple_score += 1
226 if damage == "High" and heat == "Often":
227     issues.append("CRITICAL: High damage + frequent heat use")
228     confidence += 5
229     simple_score += 2
230
231 confidence = min(confidence, 95)
232
233 # Hair Profile Scores
234 moisture = 80
235 if scalp == "Dry": moisture -= 25
236 if porosity == "High": moisture -= 20
237 if heat == "Often": moisture -= 15
238
239 if porosity == "Low": moisture += 5
240 if environment == "Dry": moisture += 10
241 moisture = max(moisture, 5)
242
243 damage_score = 0
244 if heat == "Often": damage_score += 30
245 if heat == "Sometimes": damage_score += 15
246 if chemical == "Frequently": damage_score += 30
247 if chemical == "Occasionally": damage_score += 15
248 if porosity == "High": damage_score += 10
249 if damage == "High": damage_score += 20
250 if damage == "Moderate": damage_score += 10
251 damage_score = min(damage_score, 100)
252
253 scalp_health = 85
254 if scalp == "Dry": scalp_health -= 30
255 if scalp == "Oily": scalp_health -= 20
256 if scalp == "Sensitive": scalp_health -= 15
257 if chemical == "Frequently": scalp_health -= 15
258 scalp_health = max(scalp_health, 5)
259
260 breakage = 0
261 if density == "Fine": breakage += 30
262 if porosity == "High": breakage += 25
263 if chemical == "Frequently": breakage += 25
264 if heat == "Often": breakage += 20
265 if damage == "High": breakage += 15
266 breakage = min(breakage, 100)
267
268 hair_profile = {
269     "Moisture Level": moisture,
270     "Damage Level": damage_score,
271     "Scalp Health": scalp_health,
272     "Breakage Risk": breakage
273 }
274
275 issues_str = (" ".join(issues) if issues
276               else "No major issues detected")
277
278 return issues, issues_str, confidence, simple_score, hair_profile
279
280 # =====
281 # HAIR PROFILE DASHBOARD
282 # =====
283
284 def display_hair_profile(profile):
285     st.subheader("Hair Profile Score Dashboard")
286     st.caption("Generated by Mini Brain before AI processing")
287     col1, col2, col3, col4 = st.columns(4)
288     cols = [col1, col2, col3, col4]
289     icons = {
290         "Moisture Level": ("Higher = better"),
291         "Damage Level": ("Lower = better"),
292         "Scalp Health": ("Higher = better"),
293         "Breakage Risk": ("Lower = better"),
294     }
295     for col, (label, value) in zip(cols, profile.items()):
296         note = icons[label]
297         with col:
298             st.metric(label=label, value=f"{value}/100")
299             st.progress(value / 100)
300             st.caption(note)
301
302 # =====
303 # STRUCTURED OUTPUT
304 # =====
305
306 def display_structured_output(recommendation_text):
307     sections = [
308         ("1. Recommended Shampoo Type",
309          "1. Recommended Shampoo Type"),
310         ("2. Recommended Conditioner Type",
311          "2. Recommended Conditioner Type"),
312         ("3. Recommended Treatment",
313          "3. Recommended Treatment"),
314         ("4. Suggested Weekly Routine",
315          "4. Suggested Weekly Routine"),
316         ("5. Explanation for Recommendations",
317          "5. Explanation for Recommendations"),
318     ]
319     text = recommendation_text
320     extracted = {}
321     for i, (key, _) in enumerate(sections):
322         if key in text:

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319         after = text.split(key, 1)[1]
320         end = len(after)
321         for j, (next_key, _) in enumerate(sections):
322             if j != i and next_key in after:
323                 idx = after.find(next_key)
324                 if idx < end:
325                     end = idx
326             extracted[key] = after[:end].strip()
327         if not extracted:
328             st.write(recommendation_text)
329             return
330         for key, display_title in sections:
331             if key in extracted:
332                 with st.expander(display_title, expanded=True):
333                     st.write(extracted[key])
334
335 # =====
336 # MAIN BUTTON
337 # =====
338 if st.button("Get Recommendation"):
339     detected_issues_simple, confidence_simple =
340         detect_hair_issues(
341             scalp_condition, heat_usage,
342             chemical_treatments, porosity, density
343         )
344     issues_list, issues_str, confidence, simple_score,
345     hair_profile = run_mini_brain(
346         scalp_condition, heat_usage, chemical_treatments,
347         porosity, density, hair_type, hair_goal,
348         damage_level, environment
349     )
350     st.header("Your Personalized Recommendation")
351     display_hair_profile(hair_profile)
352     st.divider()
353
354     st.subheader("Hair Health Score")
355     st.write(f"Score: {simple_score} -- Higher = more
356             attention needed")
357     if simple_score >= 8:
358         st.error("High attention needed")
359     elif simple_score >= 4:
360         st.warning("Moderate condition")
361     else:
362         st.success("Low concern -- maintenance advice")
363     st.divider()
364
365     st.subheader("Detected Issues")
366     st.write(detected_issues_simple)
367     st.divider()
368
369     st.subheader("Mini Brain -- Detailed Issue Breakdown")
370     st.caption("Rule-based analysis completed before AI
371             processing")
372     if issues_list:
373         for issue in issues_list:
374             if "RISK" in issue or "SEVERE" in issue or "
375             CRITICAL" in issue:
376                 st.error(issue)
377             else:
378                 st.warning(issue)
379     else:
380         st.success("No major issues detected")
381     st.divider()
382
383     prompt = f"""
384     You are a professional licensed trichologist and hair
385     care expert with 15+ years of experience.
386
387     User Profile:
388     - Hair Type: {hair_type}
389     - Scalp Condition: {scalp_condition}
390     - Porosity: {porosity}
391     - Density: {density}
392     - Hair Length: {hair_length}
393     - Damage Level: {damage_level}
394     - Main Goal: {hair_goal}
395     - Heat Styling: {heat_usage}
396     - Chemical Treatments: {chemical_treatments}
397     - Wash Frequency: {wash_frequency}
398     - Styling Habits: {styling_habits}
399     - Environment: {environment}
400     - Diet & Water Intake: {diet_water}
401
402     Extra Info: {user_description if user_description else "
403             None"}
404
405     System Detected Issues: {issues_str}
406     Hair Health Score: {simple_score}
407     Confidence Level: {confidence}%
408
409     Provide a recommendation using EXACTLY these sections:
410     1. Recommended Shampoo Type:
411     2. Recommended Conditioner Type:
412     3. Recommended Treatment:
413     4. Suggested Weekly Routine:
414     5. Explanation for Recommendations:
415     """
416
417     client = openai.OpenAI(api_key=api_key)
418     response = client.chat.completions.create(
419         model="gpt-4o-mini",
420         messages=[{"role": "user", "content": prompt}]
421     )
422     recommendation = response.choices[0].message.content
423
424     st.subheader("AI Recommendation")
425     display_structured_output(recommendation)
426     st.divider()
427
428     st.progress(confidence / 100)
429     st.caption(f"System Confidence Level: {confidence}%")
430     st.divider()
431
432     st.subheader("Was this recommendation helpful?")
433     rating = st.slider("Rate (1=Not helpful, 5=Excellent)",
434                       1, 5, 4)
435     if st.button("Submit Feedback"):
436         st.success("Thank you for your feedback!")
437
438 # =====
439 # SIDEBAR
440 # =====
441 st.sidebar.title("System Architecture")
442 st.sidebar.write("""
443 This prototype demonstrates a Decision Support System.
444
445 Process:
446 1 - User answers hair care survey
447 2 - Streamlit collects inputs
448 3 - Mini Brain v2.0 detects basic issues
449 4 - Enhanced Mini Brain calculates weighted scores
450 5 - Combined condition flags are triggered
451 6 - Hair Profile Score Dashboard is generated
452 7 - Hair Health Score is calculated
453 8 - Python builds a structured AI prompt
454 9 - OpenAI model analyzes hair characteristics
455 10 - AI generates personalized recommendations
456 11 - Results displayed in structured sections
457 12 - User rates recommendation (feedback loop)
458 """)
459
460 st.sidebar.title("System Features")
461 st.sidebar.write("""
462 v4.0 -- All Features Combined:
463 - Expanded questionnaire (13 questions)
464 - Hair Profile Score Dashboard
465 - Hair Health Score (simple number)
466 - Mini Brain with Confidence Score
467 - Enhanced Mini Brain (weighted + combined flags)
468 - Structured Output (5 clear sections)
469 - Day-by-day Weekly Routine
470 - Explanation broken down by hair factor
471 - User Feedback Loop
472 - Improved Prompt Engineering
473 """)
474
475 st.sidebar.info(
476     "This is a hybrid Decision Support System (Rules + AI)"
477 )

```

Listing 1. Final app.py — AI Hair Care Recommendation System v4.0

Note: The API key shown in the repository version has been replaced with a placeholder. In deployment, the key should be stored securely using environment variables or a secrets

management tool, not hardcoded in the source file.

XIV. EVALUATION

We tested the system over 13 days and focused on three things:

- **Personalization:** We tried a bunch of different combinations of inputs and checked whether the output actually changed in a meaningful way. Someone with curly, dry hair got completely different suggestions than someone with straight, oily hair — which is exactly what we were going for.
- **Explainability:** Every recommendation came with a reason. For example, the system would say something like “Hyaluronic acid is good for fine curls because it adds moisture without weighing the hair down.” That is way more useful than just getting a list of ingredients with no context.
- **Usability:** The app was easy to navigate. The five-section output format made the recommendations feel organized and approachable, and the sidebar explanation helped users understand how the system actually worked.

Accuracy, Response Time, Usability Testing:

Accuracy: The system was tested across 20 different input combinations over the 13-day evaluation period. In all cases, the AI produced recommendations that were logically consistent with the inputs provided — users with dry, curly hair consistently received moisture-focused routines while users with oily scalps received clarifying-focused advice. Because there is no labeled ground truth dataset for personalized hair care recommendations, we evaluated accuracy qualitatively by checking whether the output matched what a knowledgeable hair care advisor would reasonably suggest for each profile. All 20 test cases produced relevant, on-topic recommendations with no hallucinated or contradictory advice detected.

Response Time: API response times were measured across 15 test runs. The average time from survey submission to full recommendation display was approximately 4.2 seconds, with a minimum of 2.8 seconds and a maximum of 6.1 seconds. Response time varied slightly depending on network conditions and the length of the user’s optional text description. All responses were considered acceptable for a conversational AI tool.

Usability Testing: The app was informally tested by 5 users with varying levels of tech experience. All 5 were able to complete the survey and receive a recommendation without any assistance. Common feedback included that the five-section output format was easy to follow and that the explanation section was the most useful part. One user noted they wished they could save their results, which has been identified as a future improvement.

Comparative Analysis: Our AI Hair Assistant provides a service to users that is typically provided in-person at a hair salon or by a hair specialist, so it is difficult to present a direct comparative analysis. What can be noted is that the AI Hair Assistant does make personalized hair recommendations more accessible for end users who may not have a salon within a

reasonable distance of them or those who may not be able to afford a salon’s service price.

In contrast to the accessibility, the ability to visit a hair stylist in-person allows for a live demonstration of hair styling techniques that the AI Hair Assistant can only describe. The user may take further steps to find an online video or more in depth description of the technique, but a hair stylist could simply explain the procedure while actively performing it. Another factor to consider in favor of using an in-person hair stylist is that the end user may incorrectly self-report issues about their hair symptoms. There is room in the hair questionnaire for the user to select the incorrect hair type, texture, porosity, and so on which would then cause an irrelevant response. A licensed hair stylist should have a smaller margin of error when recognizing issues and/or qualities about a client’s hair.

Some things we know the system still cannot do: it does not recommend specific brands, it does not remember anything between sessions, and it depends on the OpenAI API being available. Those are all things we would want to improve in a future version.

XV. CONCLUSION AND FUTURE WORK

We built a working AI system that gives people personalized hair care advice based on a short survey. The combination of the Mini Brain module and GPT-4o-mini gave us something that is both reliable and flexible — it handles simple cases quickly and can still reason through complicated hair situations that a rule-based system would struggle with.

Phase 1 was about identifying the problem, doing the research, and getting a first version of the app working. Phase 2 was about formalizing the design with UML diagrams, improving the prototype, and actually evaluating whether it worked.

There is definitely more we want to do with this. Some ideas for where to take it next:

- 1) Connect it to a real product database so it can recommend specific brands, not just product types
- 2) Add the ability to upload a photo of your hair so the AI can visually analyze it
- 3) Save user sessions so the system can track progress and adjust recommendations over time
- 4) Run a formal user study to get real feedback from people with different hair types

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